University of Exeter

MTH1002

Mathematical Methods

On Campus Exam	Closed Book
Additional Materials: None	**Permitted Materials:** Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

-The **duration** of this paper is 2 hours
-Word Count: No Word Count
-Write all answers in the answer booklet provided.
-Do not open the exam paper until instructed by the Invigilator.
-Use only black pen or pencil.
-Mobile and electronic devices are not allowed.
-Do not communicate with other candidates during the exam.

Exam Instructions:

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline.

SECTION A

1. (a) The points A, B, and C have position vectors $\vec{a} = 4\hat{i} - 6\hat{j} + 5\hat{k}$, $\vec{b} = -3\hat{i} + 7\hat{k}$ and $\vec{c} = \hat{j} - 2\hat{k}$, respectively. Find

   (i) $2\vec{a} - 5\vec{b} - 2\vec{c}$

   (ii) $\overrightarrow{AC} + \overrightarrow{AB}$

   (iii) $\vec{a} \cdot \vec{c}$

   (iv) θ, where θ is the angle between $\vec{a}$ and $\vec{c}$

   (v) $\vec{a} \times \vec{c}$ (5)

   (b) Show that the line $dx + ey + f = 0$ is orthogonal to the vector $\overrightarrow{n} = (d, e)$. (5)

   (c) Use differentiation from first principles to show that $f(x) = x^{1/3}$ is not differentiable at $x = 0$. (10)

   (d) Determine the radius of convergence of the following power series.

   (i) $\sum_{k=7}^{\infty} \frac{x^k}{k^7}$

   (ii) $\sum_{k=0}^{\infty} \frac{k^k x^k}{k!}$ (10)

   (e) Use convergence tests to determine which of the following series converge.

   (i) $\sum_{n=1}^{\infty} \left(n + \frac{1}{n}\right)^3 - n^3$

   (ii) $\sum_{n=1}^{\infty} \frac{1}{2^n + 1}$ (10)

   (f) Find all complex roots of the following equation

$$z^6 - 3z^3 + 2 = 0.$$

(10)

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SECTION B

2. (a) Determine the following limits if they exist. If they do not exist, find the left-hand and right-hand limits.

   (i) $\lim_{x\to 1} \frac{x^3-1}{x-1}$,

   (ii) $\lim_{x\to 0} \frac{x-\frac{2}{x}}{x+\frac{2}{x}}$,

   (iii) $\lim_{x\to 0} \frac{e^x-1}{|x|}$. (12)

   (b) Prove that the scalar triple product is cyclic, meaning that

   $$\vec{u}\cdot(\vec{v}\times\vec{w}) = \vec{v}\cdot(\vec{w}\times\vec{u}) = \vec{w}\cdot(\vec{u}\times\vec{v}).$$

   (13)

   [**25**]

3. (a) Describe briefly the graphs of odd and even functions. (2)

   (b) Let $f : \mathbb{R} \to \mathbb{R}$ be an odd function and $g : \mathbb{R} \to \mathbb{R}$ be an even function. For all such f and g, determine whether each of the following is even or odd or neither

   (i) fg,

   (ii) $f + g$. (8)

   (c) Consider the following function

   $$f : \mathbb{R} \to \mathbb{R}, f(x) = 3x^{5/3} - 15x^{2/3}.$$

   (i) Give the definition of a critical point.

   (ii) Use this definition to find any critical points of f, and classify them.

   (iii) Sketch the graph of f. (15)

   [**25**]

4. (a) For each of the following function rules, find the Maclaurin series of order n expressing your answer using summation notation.

   (i) $f(x) = e^{1+x}$

   (ii) $f(x) = \frac{1}{2-x}$ (6)

   (b) State Taylor's Remainder Theorem and use it to approximate the error for the series of order n for the function rules given in part 4(a). (7)

   (c) The complex sine function is defined as

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

   Find all solutions of $\sin z = i$. (12)

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[END OF PAPER]